ZHAOHUI LI

Phone: (+1)-814-826-9537 | Email: nkzhaohuilee@gmail.com | Website: zhaohuilee.com

Employment

- **College of AI, Cyber and Computing, UT San Antonio** *Sep 2026 – Present*
  Postdoctoral Associate | *Mentor: Dr. Jinjun Xiong* San Antonio, TX
- **Institute for Artificial Intelligence and Data Science (IAD), University at Buffalo** *Aug 2024 – Aug 2026*
  Postdoctoral Associate | *Mentor: Dr. Jinjun Xiong* Buffalo, NY
  - AI Team Lead at the **National AI Institute for Exceptional Education (AI4ExceptionalEd)**.
  - AI Team Lead at the **Center for Early Literacy and Responsible AI (CELaRAI)**.

Education

- **The Pennsylvania State University** *Aug 2019 – May 2024*
  Ph.D. in Computer Science — Natural Language Processing Lab State College, PA
  - **Advisor:** Dr. *Rebecca J. Passonneau*
  - **Dissertation:** *Automated and Semi-Automated Assessment of STEM Reasoning Questions (**Best Doctoral Dissertation Award**)*
- **Nankai University** *Aug 2012 – Jun 2016*
  B.S. in Computer Science and Engineering Tianjin, China
  - **Advisor:** *Dr. Jun Xu* & *Dr. Jie Liu*
  - **Thesis:** BDA: Big Data Analysis as a Service (2k+ GitHub Stars)

Grants Experience

- **Organization for Autism Research (OAR) Research Grant ($50,000)** *Feb 2026 – Dec 2027*
  ***Co-PI**; Study of Parent's AI Coach for Parent-Implemented Communication Interventions for Children with ASD.*
- **Spencer Foundation — Large Research Grant on Education ($125,000–$500,000)** *July 2025 – TBD*
  ***Co-PI**; Topic: AI-Automated Behavioral Support Plans for Young Children with Autism Spectrum Disorder (In submission)*
- **NSF #2229873 — AI Institute for Exceptional Education ($20,000,000)** *Jan 2023 – Dec 2027*
  Tech Lead supporting Dr. Jinjun Xiong (Co-PI); advancing AI solutions for challenges in special education.
- **IES #R305C240046 — Center for Early Literacy and Responsible AI ($10,000,000)** *Sep 2024 – Aug 2029*
  Tech Lead supporting Dr. Jinjun Xiong (Co-PI); developing responsible AI tools for early literacy and diverse learners.
- **NSF IUSE #2236150 — Project CLASSIFIES ($300,000)** *Mar 2023 – Feb 2025*
  Contributor; developed NLP algorithms and helped write technical proposal with Dr. Rebecca Passonneau (Co-PI)
- **Penn State AI Challenge Grant — Nittany AI Challenge 2023 ($5,000)** *Jan 2023 – Jan 2024*
  ***PI**; Topic: Developed AI-driven Interactive E-Books System for Children.*

Teaching Experience

- **NSF Argumentation Workshop (PACE Lab)** *Feb 2022*
  Invited Speaker Illinois State University
  - Delivered a workshop talk on AI-assisted assessment and Large Language Models.
- **Pennsylvania State University** *Jan 2020 – May 2020*
  Teaching Assistant — Natural Language Processing (CSE597, graduate level) State College, PA
  - Teaching and grading; supported student projects.
- **Pennsylvania State University** *Aug 2019 – Dec 2019*
  Teaching Assistant — Artificial Intelligence (CMPSC442, undergraducate level) State College, PA
  - Teaching and grading; exams for 100+ students.

Service Experience

- **Proposal Reviewer:** NSF Panelist on DRL
- **Conference Reviewer:** ACL Rolling Review(ACL, EMNLP, NAACL), AAAI, AIED, SIGIR
- **Session Chair:** AIED 2025, 2026 (Language Learning & Story Generation, AI Agents for Education)
- **Program Committee Member:** AIED 2025, AIED 2026, LAK 2027
- **Student Voluteer:** AAAI 2019

Research Interests

My research focuses on developing artificial intelligence (AI), particularly multimodal large language models (LLMs), to support teaching, learning, and human connection. My work bridges AI and education by translating educational goals into task-specific computational goals. I have developed algorithms, benchmark datasets, and systems for personalized learning, including automated assessment and feedback in special education, instructional material evaluation, storybook generation, and student behavior analysis. My career goal is to develop reliable and responsible AI frameworks to promote equitable, high-quality education worldwide.

My future research will continue pursuing interdisciplinary collaborations to develop AI applications that address specific educational needs. Building on these collaborations, I will develop a theoretical framework for translating pedagogical goals into computational tasks and an agent-based system to automate parts of task formulation and evaluation design. I will also investigate how to use AI responsibly in education, focusing on potential risks such as emotional overreliance on AI and declines in students' independent reasoning. These directions will contribute to the design of education colloquim that supports learning and human connection in the ear of AI.

Keywords: Large Language Models, Natural Language Processing, Educational Assessment, Educational Technology, Early Literacy, Learning Science, AI Literacy, Art Education, Special Education

Publications

[1] Akemoğlu, Y., **Li, Z.**, Singh, S. P., Zheng, Q., Roberts, E. B., & Xiong, J. Parent's AI Coach (PaiCoach): A Single-Case Experimental Study of an AI-Supported Parent Coaching System for Autistic Children. *Journal of Autism and Developmental Disorders*.

[2] Xiao, F., **Li, Z.**, Jiang, S., & Xiong, J. (2026). Design and Feasibility of an LLM-Powered Humanoid Robot as a Reading Companion to Support Children's AI Literacy. In *Proceedings of the International Conference on Artificial Intelligence in Education*. (AIED 2026)

[3] **Li, Z.**, He, P., Chen, Z., Liu, H., Wang, Z., Li, T., & Xiong, J. (2026). SciEval: A Benchmark for Automatic Evaluation of K–12 Science Instructional Materials. In *Proceedings of the International Conference on Artificial Intelligence in Education*. (AIED 2026)

[4] **Li, Z.**, Li, N., & Xiong, J. (2026). LLM4Art: Using Large Language Models for Art Interpretation. In *Proceedings of the International Conference on Learning Analytics & Knowledge*. (LAK 2026)

[5] He, P., **Li, Z.**, Wang, Z., Xiong, J., & Li, T. (2026). Judging the Judges: Human Validation of Multi-LLM Evaluation for High-Quality K–12 Science Instructional Materials. In *Proceedings of the Computing Conference*. (Computing Conference 2026)

[6] **Li, Z.**, Zheng, Q., Liu, D., Christ, T. M., Strong, J. Z., Tortorelli, L. S., Wang, X. C., Alwan, A., Hakkani Tur, D., & Xiong, J. (2026, April). AIRE: AI Reading Enhancer for Personalized Decodable Texts in Early Literacy. Poster presented at the *American Educational Research Association Annual Meeting*, Los Angeles, CA. (AERA 2026)

[7] **Li, Z.**; Xiao, F.; Lin, J.; Zou, X.; Zheng, Q.; Xiong, J. (2025, July). StoryLab: Empowering Personalized Learning for Children through Teacher-Guided Multimodal Story Generation. In *International Conference on Artificial Intelligence in Education* (pp. 285–292). Cham: Springer Nature Switzerland. (AIED 2025)

[8] **Li, Z.**, Akemoglu, Y., Lyu, J., Zheng, Q., & Xiong, J. (2025, July). ASD-HI: A Parent-Child Interaction Dataset for Automated Assessment of Home Intervention. In *International Conference on Artificial Intelligence in Education* (pp. 48–62). Cham: Springer Nature Switzerland. (AIED 2025)

[9] Xiao, F., **Li, Z.**, Lin, J., Zou, X., Yang, D., Zou, E. W., & Xiong, J. (2025). Leveraging an LLM-enhanced bilingual conversational agent for EFL children's dialogic reading: Insights from children, parents, and educators. *Computers and Education: Artificial Intelligence*, 100484.

[10] Xiao, F., Zou, E. W., Lin, J., **Li, Z.**, & Yang, D. (2025). Parent-led vs. AI-guided dialogic reading: Evidence from a randomized controlled trial in children's e-book context. *British Journal of Educational Technology*, 56(5), 1784–1813.

[11] Sheikhi_Karizaki, M., **Li, Z.**, & Passonneau, R. J. (2025, July). Criteria for Demonstration Examples for LLM Assessment of Science Essays. In *Workshop on Epistemics and Decision-Making in AI-Supported Education, Annual Conference on Artificial Intelligence in Education*. (AIED 2025)

[12] **Li, Z.**, & Passonneau, R. J. (2024). Joint Training for Selective Prediction. *arXiv preprint* arXiv:2410.24029.

[13] **Li, Z.**, Lloyd, S. E., Beckman, M. D., & Passonneau, R. J. (2023, December). Answer-state recurrent relational network (AsRRN) for constructed response assessment and feedback grouping. In *The 2023 Conference on Empirical Methods in Natural Language Processing*. (EMNLP 2023)

[14] **Li, Z.**, Zhang, C., Jin, Y., Cang, X., Puntambekar, S., & Passonneau, R. J. (2023, June). Learning when to defer to humans for short answer grading. In *International Conference on Artificial Intelligence in Education* (pp. 414–425). Cham: Springer Nature Switzerland. (AIED 2023)

[15] **Li, Z.**; Lloyd, S.; Beckman, M. D.; Passonneau, R. J. (2023). *ISTUDIO: A Statistical Automatic Response Assessment Dataset*. Penn State Data Commons.

[16] Passonneau, R., Koenig, K., **Li, Z.**, & Soddano, J. (2023). The ideal versus the real deal in assessment of physics lab report writing. *European Journal of Applied Sciences*, 11(2).

[17] Lloyd, S., Beckman, M. D., Pearl, D. K., Passonneau, R. J., **Li, Z.**, & Wang, Z. (2022, September). Foundations for AI-Assisted Formative Assessment Feedback for Short-Answer Tasks in Large-Enrollment Classes. In *Proceedings of the 11th International Conference on Teaching Statistics (ICOTS 11)*. DOI: 10.52041/iase.icots11.T3C3.

[18] **Li, Z.**; Atil, B.; Koenig, K.; Passonneau, R. J. (2022). *Reliable Rubric-Based Assessment of Physics Lab Reports: Data for Machine Learning*. Penn State Data Commons.

[19] **Li, Z.**, Tomar, Y., & Passonneau, R. J. (2021, November). A semantic feature-wise transformation relation network for automatic short answer grading. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* (pp. 6030–6040). (EMNLP 2021)

[20] **Li, Z.**, Feng, Y., Xu, J., Guo, J., Lan, Y., & Cheng, X. (2019, July). Teaching machines to extract main content for machine reading comprehension. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 33, No. 1, pp. 9973–9974).

[21] **Li, Z.**, Xu, J., Lan, Y., Guo, J., Feng, Y., & Cheng, X. (2018, September). Hierarchical answer selection framework for multi-passage machine reading comprehension. In *China Conference on Information Retrieval* (pp. 93–104). Cham: Springer International Publishing. (**Best Paper Nominee**)

[22] Guo, T., Xu, J., Yan, X., Hou, J., Li, P., **Li, Z.**, . . . & Cheng, X. (2016, October). Ease the process of machine learning with dataflow. In *Proceedings of the 25th ACM International Conference on Information and Knowledge Management (CIKM 2016)* (pp. 2437–2440). (**Best Demo Paper Nominee**)

[23] Wang, Y.; **Li, Z.**; Liu, J.; He, Z.; Huang, Y.; Li, D. (2014). *Word Vector Modeling for Sentiment Analysis of Product Reviews*. NLPCC 2014.

Ongoing Research Projects

- **Automated Assessment of an ASD Home-Based Intervention** *Nov 2024 – Present*
  Research Lead; work with Prof. Jinjun Xiong (Co-PI), National AI Institute for Exceptional Education, University at Buffalo.
  - Lead the creation of a large-scale parent–child interaction dataset for automated assessment of home-based ASD interventions.
  - Design and implement an LLM-based automatic evaluation pipeline for behavioral and communicative outcomes.
  - Fine-tune large language models for speech recognition, visual information processing, and parent-facing feedback generation.
  - Develop and deploy the PiCoach platform; recruit parents and children to conduct user studies and collect real-world intervention data.

- **Innovating Beginning Reading Instruction for Culturally and Linguistically Diverse Learners** *Jan 2025 – Present*
  AI Tech Lead; work with Prof. X. Christine Wang (PI), Center for Early Literacy and Responsible AI, University at Buffalo.
  - Lead the development of the AI Reading Enhancer platform for early literacy instruction.
  - Design and implement a personalized decodable book generation pipeline aligned with K-2 child reading practices.
  - Fine-tune large language models for storybook text generation and illustration content creation.
  - Co-design and validate decodable storybook evaluation rubrics in collaboration with early literacy experts.

- **Exploring LLM-Based Embodied Agents for Early Childhood AI Literacy** *Nov 2025 – Present*
  Research Lead; collaboration with Prof. Shiyang Jiang (Graduate School of Education, University of Pennsylvania).
  - Fine-tune and deploy multimodal large language models for embodied robotic agents supporting K–2 AI literacy.
  - Evaluate and select child-appropriate robotic platforms for early childhood educational settings.
  - Collect and analyze multimodal home-environment interaction data to iteratively improve model performance.
  - Lead the development of an NSF proposal focused on AI literacy, embodied learning, computation thinking, and responsible AI for young learners.

- **Automated Evaluation of Instructional Materials for K–12 Science Education** *July 2025 – Present*
  Research Lead; collaboration with Prof. Tingting Li and Prof. Peng He (College of Education, Washington State University).
  - Design and implement an evaluation framework for assessing large language models on instructional material analysis and quality judgment tasks.
  - Recruit, train, and calibrate human annotators to construct a high-quality lesson and instructional-material dataset based on OpenSciEd curricula.
  - Fine-tune Qwen-Omni-3 models to generate evidence-based assessments, pedagogical reasoning, and actionable suggestions for instructional materials.
  - Lead the drafting and submission of NSF proposal related to AI-assisted instructional evaluation and science education.

- **LLM4Art: Using Large Language Models for Art Interpretation** *July 2025 – Present*
  Research Lead; collaboration with Prof. Kyunghee Pyun (Fashion Institute of Technology, State University of New York).
  - Design and implement an evaluation framework for assessing large language models on art interpretation tasks.
  - Develop a prototype tool to support art interpretation and art education using LLM-based assistance.
  - Co-curate and annotate a large-scale Art + LLM dataset for model training and evaluation.
  - Train and fine-tune art-specialized LLMs for interpretive reasoning and educational applications.

- **LLM4Simulation: Using Large Language Models for Dyslexia Simulations** *Jan 2025 – Present*
  Research Lead; collaboration with Prof. Xinxia Li (School of Education and Human Development, George Washington University).
  - Conduct a systematic review of the design, delivery, and evaluation of dyslexia simulation technologies.
  - Develop an LLM-based prototype to generate interactive activities for dyslexia simulations supporting public awareness and special education training.
  - Co-design immersive simulation environments integrating virtual reality (VR) and LLM-driven agents.
  - Lead the drafting and submission of NSF proposal related to AI-driven disability simulations and inclusive education.